

An Analysis of Prompt engineering

Prompt engineering

Prompt engineering -- Part 1

Rationale Research has found that the performance of large language models (LLMs) is highly sensitive to choices such as the ordering of examples, the quality of demonstration labels, and even small variations in phrasing. In some cases, reordering examples in a prompt produced accuracy shifts of more than 40 percent.

Prompt engineering -- Part 2

Multi-shot A prompt may include a few examples for a model to learn from in context, an approach called few-shot learning. For example, the prompt may ask the model to complete "maison → house, chat → cat, chien →", with the expected response being dog.

Prompt engineering -- Part 3

Prompting to estimate model sensitivity Research consistently demonstrates that LLMs are highly sensitive to subtle variations in prompt formatting, structure, and linguistic properties. Some studies have shown up to 76 accuracy points across formatting changes in few-shot settings. Linguistic features significantly influence prompt effectiveness—such as morphology, syntax, and lexico-semantic changes—which meaningfully enhance task performance across a variety of tasks. Clausal syntax, for example, improves consistency and reduces uncertainty in knowledge retrieval. This sensitivity persists even with larger model sizes, additional few-shot examples, or instruction tuning. To address sensitivity of models and make them more robust, several evaluative methods have been proposed. FormatSpread facilitates systematic analysis by evaluating a range of plausible prompt formats, offering a more comprehensive performance interval. Similarly, PromptEval estimates performance distributions across diverse prompts, enabling robust metrics such as performance quantiles and accurate evaluations under constrained budgets.

Prompt engineering -- Part 4

Retrieval-augmented generation (RAG) Retrieval-augmented generation is a technique that enables GenAI models to retrieve and incorporate new information. It modifies interactions with an LLM so that the model responds to user queries with reference to a specified set of documents, using this information to supplement information from its pre-existing training data. This allows LLMs to use domain-specific and/or updated information. RAG improves large language models by incorporating information retrieval before generating responses. Unlike traditional LLMs that rely on static training data, RAG pulls

relevant text from databases, uploaded documents, or web sources. By dynamically retrieving information, RAG enables AI to generate more accurate responses and fewer AI hallucinations without frequent retraining.

Prompt engineering -- Part 5

Image prompting In 2023, Meta's AI research released Segment Anything, a computer vision model that can perform image segmentation by prompting. As an alternative to text prompts, Segment Anything can accept bounding boxes, segmentation masks, and foreground/background points.

Prompt engineering -- Part 6

Prompt engineering is the process of structuring natural language inputs (known as prompts) to produce specified outputs from a generative artificial intelligence (GenAI) model. Context engineering is the related area of software engineering that focuses on the management of non-prompt contexts supplied to the GenAI model, such as metadata, API tools, and tokens. It can also be defined as the practice of designing and refining input instructions given to a generative AI model to produce more accurate, relevant, or useful outputs. Effective prompt engineering involves understanding how a model interprets language, and may include techniques such as few-shot prompting, chain-of-thought prompting, and role assignment. It is increasingly considered a skill for working with large language models (LLMs) in both research and professional contexts. During the 2020s AI boom, prompt engineering became regarded as a business capability across corporations and industries. Employees with the title prompt engineer were hired to create prompts that would increase productivity and efficacy, although the individual title has since lost traction amid AI models that produce better prompts than humans and corporate training in prompting for general employees. Common prompting techniques include multi-shot, chain-of-thought, and tree-of-thought prompting, as well as the use of assigning roles to the model. Automated prompt generation methods, such as retrieval-augmented generation (RAG), provide for greater accuracy and a wider scope of functions for prompt engineers. Prompt injection is a type of cybersecurity attack that targets machine learning models through malicious prompts.

Prompt engineering -- Part 7

Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

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Prompt engineering -- Part 8

There are two LLMs. One is the target LLM, and another is the prompting LLM. Prompting LLM is presented with example input-output pairs, and asked to generate instructions that could have caused a model following the instructions to generate the outputs, given the inputs. Each of the generated instructions is used to prompt the target LLM, followed by each of the inputs. The log-probabilities of the outputs are computed and added. This is the score of the instruction. The highest-scored instructions are given to the prompting LLM for further variations. Repeat until some stopping criteria is reached, then output the highest-scored instructions. CoT examples can be generated by LLM themselves. In "auto-CoT", a library of questions are converted to vectors by a model such as BERT. The question vectors are clustered. Questions close to the centroid of each cluster are selected, in order to have a subset of diverse questions. An LLM does zero-shot CoT on each selected question. The question and the corresponding CoT answer are added to a dataset of demonstrations. These diverse demonstrations can then added to prompts for few-shot learning.

safeguards and influence model behaviour. While LLMs are designed to follow trusted instructions, they can be manipulated into carrying out unintended responses through carefully crafted inputs.

Prompt engineering -- Part 9

A text-to-image prompt commonly includes a description of the subject of the art, the desired medium (such as digital painting or photography), style (such as hyperrealistic or pop-art), lighting (such as rim lighting or crepuscular rays), color, and texture. Word order also affects the output of a text-to-image prompt. Words closer to the start of a prompt may be emphasized more heavily.

Prompt engineering -- Part 10

Limitations The process of writing and refining a prompt for an LLM or generative AI shares some parallels with an iterative engineering design process, such as by discovering reusable best practices through reproducible experimentation. But the techniques that improve performance depend heavily on the specific model being used. Such patterns are also volatile and exhibit significantly different results from seemingly insignificant prompt changes.

Prompt engineering -- Part 11

Prompt injection is a cybersecurity exploit in which adversaries craft inputs that appear legitimate but are designed to cause unintended behavior in machine learning models, particularly large language models. This attack takes advantage of the model's inability to distinguish between developer-defined prompts and user inputs, allowing adversaries to bypass